

Research Notes on GPT language model

GPT language model

>> Section 1

A foundational GPT model can be further adapted to produce more targeted systems directed to specific tasks and/or subject-matter domains. Methods for such adaptation can include additional fine-tuning (beyond that done for the foundation model) as well as certain forms of prompt engineering. An important example of this is fine-tuning models to follow instructions, which is of course a fairly broad task but more targeted than a foundation model. In January 2022, OpenAI introduced "InstructGPT" – a series of models which were fine-tuned to follow instructions using a combination of supervised training and reinforcement learning from human feedback (RLHF) on base GPT-3 language models. Advantages this had over the bare foundational models included higher accuracy, less negative/toxic sentiment, and generally better alignment with user needs. Hence, OpenAI began using this as the basis for its API service offerings. Other instruction-tuned models have been released by others, including a fully open version. Another (related) kind of task-specific models are chatbots, which engage in human-like conversation. In November 2022, OpenAI launched ChatGPT – an online chat interface powered by an instruction-tuned language model trained in a similar fashion to InstructGPT. They trained this model using RLHF, with human AI trainers providing conversations in which they played both the user and the AI, and mixed this new dialogue dataset with the InstructGPT dataset for a conversational format suitable for a chatbot. Other major chatbots currently include Microsoft's Bing Chat, which uses OpenAI's GPT-4 (as part of a broader close collaboration between OpenAI and Microsoft), and Google's competing chatbot Gemini (initially based on their LaMDA family of conversation-trained language models, with plans to switch to PaLM). Yet another kind of task that a GPT can be used for is the meta-task of generating its own instructions, like developing a series of prompts for 'itself' to be able to effectuate a more general goal given by a human user. This is known as an AI agent, and more specifically a recursive one because it uses results from its previous self-instructions to help it form its subsequent prompts; the first major example of this was Auto-GPT (which uses OpenAI's GPT models), and others have since been developed as well. Chain-of-thought reasoning is a prompting technique in which a language model generates intermediate reasoning steps before arriving at a final answer. This approach has been shown to improve performance on tasks requiring multi-step reasoning, such as mathematical problem solving and logical inference. By producing step-by-step explanations, the model is able to decompose complex problems into smaller, more manageable parts. This technique is often used in

conjunction with few-shot prompting, where examples demonstrating the reasoning process are included in the input.

>> Section 2

Background During the 2010s, improved machine learning algorithms, more powerful computers, and an increase in the amount of digitized material allowed for an AI boom. Separately, the concept of generative pre-training (GP) was a long-established technique in machine learning. GP is a form of self-supervised learning wherein a model is first trained on a large, unlabeled dataset (the "pre-training" step) to learn to generate data points. This pre-trained model is then adapted to a specific task using a labeled dataset (the "fine-tuning" step). The transformer architecture for deep learning is the core technology of a GPT. Developed by researchers at Google, it was introduced in the paper "Attention Is All You Need", which was released on June 12, 2017. The transformer architecture solved many of the performance issues that were associated with older recurrent neural network (RNN) designs for natural language processing (NLP). The architecture's use of an attention mechanism allows models to process entire sequences of text at once, enabling the training of much larger and more sophisticated models. Since 2017, available transformer-based NLP systems have been capable of processing, mining, organizing, connecting, contrasting, and summarizing texts as well as answering questions from textual input.

>> Section 3

Scaling laws Scaling laws describe empirical relationships between the performance of large language models and factors such as model size, dataset size, and computational resources. Empirical work has found that performance tends to follow approximate power-law relationships as these factors increase. Larger models trained on more data generally achieve lower training loss and better generalization. Later work suggests that performance is not determined by parameter count alone, but by how model size, data, and compute are balanced during training. These observations have influenced the development of successive GPT models, particularly in decisions about architecture design, dataset composition, and training strategies.

>> Section 4

Foundation models A foundation model is an AI model trained on broad data at scale such that it can be adapted to a wide range of downstream tasks. The most recent OpenAI's GPT-n series model is GPT-5. Other such models include Google's PaLM, a broad foundation model

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that has been compared to GPT-3 and has been made available to developers via an API, and Together's GPT-JT, which has been reported as the closest-performing open-source alternative to GPT-3 (and is derived from earlier open-source GPTs). Meta AI (formerly Facebook) also has a generative transformer-based foundational large language model, known as LLaMA. Foundational GPTs can also employ modalities other than text, for input and/or output. GPT-4 is a multi-modal LLM that is capable of processing text and image input (though its output is limited to text). Regarding multimodal output, some generative transformer-based models are used for text-to-image technologies such as diffusion and parallel decoding. Such kinds of models can serve as visual foundation models (VFM) for developing downstream systems that can work with images.

>> Section 5

A generative pre-trained transformer (GPT) is a type of large language model (LLM) that is widely used in generative artificial intelligence chatbots. GPTs are based on a deep learning architecture called the transformer. They are pre-trained on large datasets of unlabeled content, and able to generate novel content. OpenAI was the first to apply generative pre-training to the transformer architecture, introducing the GPT-1 model in 2018. The company has since released many bigger GPT models. The chatbot ChatGPT, released in late 2022 (using GPT-3.5), was followed by many competitor chatbots using their own generative pre-trained transformers to generate text, such as Gemini, DeepSeek and Claude. GPTs are primarily used to generate text, but can be trained to generate other kinds of data. For example, GPT-4o can process and generate text, images and audio. To improve performance on complex tasks, some GPTs, such as OpenAI o3, allocate more computation time analyzing the problem before generating an output, and are called reasoning models. In 2025, GPT-5 was released with a router that automatically selects whether to use a faster model or slower reasoning model based on the provided task.